

Homework 7 Solutions: Unconstrained Optimization with Two Variables

GSE 510

Practice Problems: Work together in class

1. Consider the function

$$f(x, y) = 10 + 4x + 6y - x^2 - y^2.$$

(a) The first partial derivatives are

$$f_x = 4 - 2x$$

and

$$f_y = 6 - 2y.$$

(b) The critical point solves

$$f_x = 0, \quad f_y = 0.$$

Thus,

$$4 - 2x = 0$$

and

$$6 - 2y = 0.$$

Solving,

$$x = 2, \quad y = 3.$$

Therefore, the critical point is

$$(2, 3).$$

(c) The second partial derivatives are

$$f_{xx} = -2, \quad f_{yy} = -2, \quad f_{xy} = 0.$$

(d) The Hessian matrix is

$$H = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}.$$

(e) The determinant of the Hessian is

$$|H| = (-2)(-2) - 0(0) = 4.$$

Since

$$|H| > 0$$

and

$$f_{xx} = -2 < 0,$$

the critical point is a local maximum.

2. Consider the function

$$f(x, y) = x^2 + y^2 - 4x - 6y + 20.$$

(a) The first partial derivatives are

$$f_x = 2x - 4$$

and

$$f_y = 2y - 6.$$

(b) The critical point solves

$$2x - 4 = 0$$

and

$$2y - 6 = 0.$$

Hence,

$$x = 2, \quad y = 3.$$

The critical point is

$$(2, 3).$$

(c) The second partial derivatives are

$$f_{xx} = 2, \quad f_{yy} = 2, \quad f_{xy} = 0.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

(d) The determinant is

$$|H| = 2(2) - 0(0) = 4.$$

Since

$$|H| > 0$$

and

$$f_{xx} = 2 > 0,$$

the critical point is a local minimum.

(e) The minimum value is

$$f(2, 3) = 2^2 + 3^2 - 4(2) - 6(3) + 20.$$

Therefore,

$$f(2, 3) = 4 + 9 - 8 - 18 + 20 = 7.$$

The minimum value is

$$7.$$

3. Consider the function

$$f(x, y) = xy.$$

(a) The first partial derivatives are

$$f_x = y$$

and

$$f_y = x.$$

(b) The critical point solves

$$y = 0$$

and

$$x = 0.$$

Therefore, the critical point is

$$(0, 0).$$

(c) The second partial derivatives are

$$f_{xx} = 0, \quad f_{yy} = 0, \quad f_{xy} = 1.$$

Thus, the Hessian matrix is

$$H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

(d) The determinant of the Hessian is

$$|H| = 0(0) - 1(1) = -1.$$

(e) Since

$$|H| < 0,$$

the critical point is a saddle point. It is neither a local maximum nor a local minimum.

4. Suppose a firm's profit depends on two choice variables, labor L and capital K :

$$\pi(L, K) = 100L + 120K - 2L^2 - 3K^2 - LK.$$

(a) The first partial derivatives are

$$\pi_L = 100 - 4L - K$$

and

$$\pi_K = 120 - 6K - L.$$

(b) The first-order conditions are

$$100 - 4L - K = 0$$

and

$$120 - 6K - L = 0.$$

Rearranging,

$$4L + K = 100$$

and

$$L + 6K = 120.$$

From the first equation,

$$K = 100 - 4L.$$

Substitute into the second equation:

$$L + 6(100 - 4L) = 120.$$

Therefore,

$$L + 600 - 24L = 120,$$

so

$$-23L = -480.$$

Hence,

$$L^* = \frac{480}{23}.$$

Then

$$K^* = 100 - 4\left(\frac{480}{23}\right) = \frac{2300}{23} - \frac{1920}{23} = \frac{380}{23}.$$

Therefore, the critical point is

$$\left(\frac{480}{23}, \frac{380}{23}\right).$$

(c) The second partial derivatives are

$$\pi_{LL} = -4, \quad \pi_{KK} = -6, \quad \pi_{LK} = -1.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} -4 & -1 \\ -1 & -6 \end{bmatrix}.$$

- (d) The determinant of the Hessian is

$$|H| = (-4)(-6) - (-1)(-1) = 24 - 1 = 23.$$

Since

$$|H| > 0$$

and

$$\pi_{LL} = -4 < 0,$$

the critical point is a local maximum.

- (e) The optimal values are

$$L^* = \frac{480}{23} \quad \text{and} \quad K^* = \frac{380}{23}.$$

Economically, these are the firm's profit-maximizing choices of labor and capital, assuming the firm can choose both inputs freely.

5. Consider the function

$$f(x, y) = x^2 - y^2.$$

- (a) The first partial derivatives are

$$f_x = 2x$$

and

$$f_y = -2y.$$

- (b) The critical point solves

$$2x = 0$$

and

$$-2y = 0.$$

Therefore,

$$x = 0, \quad y = 0.$$

The critical point is

$$(0, 0).$$

- (c) The second partial derivatives are

$$f_{xx} = 2, \quad f_{yy} = -2, \quad f_{xy} = 0.$$

Thus,

$$H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

- (d) The determinant is

$$|H| = 2(-2) - 0(0) = -4.$$

Since

$$|H| < 0,$$

the critical point is a saddle point.

- (e) Intuitively, the function increases as we move away from the origin in the x direction because of the x^2 term. But it decreases as we move away from the origin in the y direction because of the $-y^2$ term. So the origin is not a maximum or a minimum; it is a saddle point.

Homework Problems: Submit individually

6. Consider the function

$$f(x, y) = 24x + 18y - 3x^2 - 2y^2.$$

(a) The first partial derivatives are

$$f_x = 24 - 6x$$

and

$$f_y = 18 - 4y.$$

(b) The critical point solves

$$24 - 6x = 0$$

and

$$18 - 4y = 0.$$

Therefore,

$$6x = 24$$

and

$$4y = 18.$$

Hence,

$$x = 4, \quad y = \frac{9}{2}.$$

The critical point is

$$\left(4, \frac{9}{2}\right).$$

(c) The second partial derivatives are

$$f_{xx} = -6, \quad f_{yy} = -4, \quad f_{xy} = 0.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} -6 & 0 \\ 0 & -4 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-6)(-4) - 0(0) = 24.$$

Since

$$|H| > 0$$

and

$$f_{xx} = -6 < 0,$$

the critical point is a local maximum.

(e) The maximum value is

$$f\left(4, \frac{9}{2}\right) = 24(4) + 18\left(\frac{9}{2}\right) - 3(4)^2 - 2\left(\frac{9}{2}\right)^2.$$

Compute each term:

$$24(4) = 96,$$

$$18\left(\frac{9}{2}\right) = 81,$$

$$3(4)^2 = 48,$$

and

$$2\left(\frac{9}{2}\right)^2 = 2\left(\frac{81}{4}\right) = \frac{81}{2}.$$

Therefore,

$$f\left(4, \frac{9}{2}\right) = 96 + 81 - 48 - \frac{81}{2}.$$

Thus,

$$f\left(4, \frac{9}{2}\right) = 129 - \frac{81}{2} = \frac{258}{2} - \frac{81}{2} = \frac{177}{2}.$$

The maximum value is

$$\frac{177}{2}.$$

7. Consider the function

$$f(x, y) = x^2 + 2y^2 + 2xy - 6x - 10y.$$

(a) The first partial derivatives are

$$f_x = 2x + 2y - 6$$

and

$$f_y = 4y + 2x - 10.$$

(b) The first-order conditions are

$$2x + 2y - 6 = 0$$

and

$$2x + 4y - 10 = 0.$$

Divide the first equation by 2:

$$x + y = 3.$$

Divide the second equation by 2:

$$x + 2y = 5.$$

Subtract the first equation from the second:

$$y = 2.$$

Then

$$x + 2 = 3,$$

so

$$x = 1.$$

The critical point is

$$(1, 2).$$

(c) The second partial derivatives are

$$f_{xx} = 2, \quad f_{yy} = 4, \quad f_{xy} = 2.$$

Therefore,

$$H = \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}.$$

(d) The determinant of the Hessian is

$$|H| = 2(4) - 2(2) = 8 - 4 = 4.$$

(e) Since

$$|H| > 0$$

and

$$f_{xx} = 2 > 0,$$

the critical point is a local minimum.

(f) The value of the function at the critical point is

$$f(1, 2) = 1^2 + 2(2)^2 + 2(1)(2) - 6(1) - 10(2).$$

Therefore,

$$f(1, 2) = 1 + 8 + 4 - 6 - 20 = -13.$$

8. Suppose a firm chooses advertising A and research spending R . Its profit is

$$\pi(A, R) = 50A + 80R - A^2 - 2R^2 - AR.$$

(a) The first partial derivatives are

$$\pi_A = 50 - 2A - R$$

and

$$\pi_R = 80 - 4R - A.$$

(b) The first-order conditions are

$$50 - 2A - R = 0$$

and

$$80 - 4R - A = 0.$$

Rearranging,

$$2A + R = 50$$

and

$$A + 4R = 80.$$

From the first equation,

$$R = 50 - 2A.$$

Substitute into the second equation:

$$A + 4(50 - 2A) = 80.$$

Therefore,

$$A + 200 - 8A = 80.$$

So

$$-7A = -120,$$

and hence

$$A^* = \frac{120}{7}.$$

Then

$$R^* = 50 - 2\left(\frac{120}{7}\right) = \frac{350}{7} - \frac{240}{7} = \frac{110}{7}.$$

Therefore, the critical point is

$$\left(\frac{120}{7}, \frac{110}{7}\right).$$

(c) The second partial derivatives are

$$\pi_{AA} = -2, \quad \pi_{RR} = -4, \quad \pi_{AR} = -1.$$

Thus,

$$H = \begin{bmatrix} -2 & -1 \\ -1 & -4 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-2)(-4) - (-1)(-1) = 8 - 1 = 7.$$

Since

$$|H| > 0$$

and

$$\pi_{AA} = -2 < 0,$$

the Hessian implies that the critical point is a local maximum.

(e) The cross-partial derivative is

$$\pi_{AR} = -1.$$

This means that increasing research spending lowers the marginal profit from advertising, and increasing advertising lowers the marginal profit from research. The two activities interact negatively in the profit function.

9. Consider the function

$$f(x, y) = x^3 - 3x + y^2.$$

(a) The first partial derivatives are

$$f_x = 3x^2 - 3$$

and

$$f_y = 2y.$$

(b) Critical points solve

$$3x^2 - 3 = 0$$

and

$$2y = 0.$$

From the first equation,

$$x^2 = 1,$$

so

$$x = -1 \quad \text{or} \quad x = 1.$$

From the second equation,

$$y = 0.$$

Therefore, the critical points are

$$(-1, 0) \quad \text{and} \quad (1, 0).$$

(c) The second partial derivatives are

$$f_{xx} = 6x, \quad f_{yy} = 2, \quad f_{xy} = 0.$$

Therefore,

$$H = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}.$$

(d) At $(-1, 0)$,

$$H(-1, 0) = \begin{bmatrix} -6 & 0 \\ 0 & 2 \end{bmatrix}.$$

The determinant is

$$|H| = (-6)(2) - 0(0) = -12.$$

Since

$$|H| < 0,$$

the point $(-1, 0)$ is a saddle point.

At $(1, 0)$,

$$H(1, 0) = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}.$$

The determinant is

$$|H| = 6(2) - 0(0) = 12.$$

Since

$$|H| > 0$$

and

$$f_{xx} = 6 > 0,$$

the point $(1, 0)$ is a local minimum.

- (e) Checking only the first-order conditions tells us where the slope is zero, but it does not tell us the shape of the function around those points. A critical point can be a maximum, minimum, or saddle point. The second-order conditions help classify the critical points.

10. **Slightly harder problem.** Suppose a firm chooses two products, q_1 and q_2 . Its profit function is

$$\pi(q_1, q_2) = 40q_1 + 50q_2 - 2q_1^2 - 3q_2^2 - q_1q_2.$$

- (a) The marginal profits are

$$\pi_{q_1} = 40 - 4q_1 - q_2$$

and

$$\pi_{q_2} = 50 - 6q_2 - q_1.$$

- (b) The first-order conditions are

$$40 - 4q_1 - q_2 = 0$$

and

$$50 - 6q_2 - q_1 = 0.$$

Rearranging,

$$4q_1 + q_2 = 40$$

and

$$q_1 + 6q_2 = 50.$$

From the first equation,

$$q_2 = 40 - 4q_1.$$

Substitute into the second equation:

$$q_1 + 6(40 - 4q_1) = 50.$$

Thus,

$$q_1 + 240 - 24q_1 = 50.$$

Therefore,

$$-23q_1 = -190.$$

Hence,

$$q_1^* = \frac{190}{23}.$$

Then

$$q_2^* = 40 - 4\left(\frac{190}{23}\right) = \frac{920}{23} - \frac{760}{23} = \frac{160}{23}.$$

The critical point is

$$\left(\frac{190}{23}, \frac{160}{23}\right).$$

(c) The second partial derivatives are

$$\pi_{q_1 q_1} = -4, \quad \pi_{q_2 q_2} = -6, \quad \pi_{q_1 q_2} = -1.$$

Therefore,

$$H = \begin{bmatrix} -4 & -1 \\ -1 & -6 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-4)(-6) - (-1)(-1) = 24 - 1 = 23.$$

Since

$$|H| > 0$$

and

$$\pi_{q_1 q_1} = -4 < 0,$$

the critical point is a local maximum.

(e) The firm's maximum profit is

$$\pi\left(\frac{190}{23}, \frac{160}{23}\right).$$

Substituting,

$$\pi = 40\left(\frac{190}{23}\right) + 50\left(\frac{160}{23}\right) - 2\left(\frac{190}{23}\right)^2 - 3\left(\frac{160}{23}\right)^2 - \left(\frac{190}{23}\right)\left(\frac{160}{23}\right).$$

This simplifies to

$$\pi^* = \frac{139000}{529}.$$

Since

$$\frac{139000}{529} \approx 262.76,$$

the maximum profit is approximately

$$262.76.$$

(f) The cross-partial derivative is

$$\pi_{q_1 q_2} = -1.$$

This means that increasing production of product 2 reduces the marginal profit from product 1, and increasing production of product 1 reduces the marginal profit from product 2. Economically, the two products interact negatively in production or profit, perhaps because they compete for shared resources.